

Theorem 19.1 Let $A \in \mathbb{C}^{n \times n}$ such that A has characteristic and minimal polynomial of
 $p(\lambda) = \prod_{\sum n_i = n} (\lambda_i - \lambda)^{n_i}$ and $\mu = \prod_i (x - \lambda_i)^{m_i}$ ($m_i \leq n_i$)
 ($d_i \leq n_i$)

Also assuming that nullity $(A - \lambda_i I) = d_i$ we have that

- i. The Jordan form J consists of $\sum d_i$ blocks (one block for each e-vector)
- ii. m_i is the size of the largest Jordan block associated to λ_i

Observation 1: If A has e-values of $\lambda_1, \dots, \lambda_\ell$ where $\lambda_i \neq \lambda_j$ then $\mathbb{C}^n = \bigoplus_{i=1}^{\ell} \text{null}(A - \lambda_i I)^{\alpha_i}$ for some α_i .

Observation 2: $\text{null}(A - \lambda_i I)^{\alpha_i}$ is an A -invariant subspace for some α_i ($\alpha_i \leq m_i$)

Observation 3: If J is an $\ell \times \ell$ Jordan block

matrix associated to λ , then $(J - \lambda I)^l = 0$
but $(J - \lambda I)^k \neq 0$ for any $k < l$.

$$J = \begin{bmatrix} \lambda & 1 \\ & \lambda \\ & & \lambda \end{bmatrix} \rightarrow J - \lambda I = \begin{bmatrix} & 1 \\ & \\ & & \end{bmatrix}$$

$$\hookrightarrow (J - \lambda I)^2 = \begin{bmatrix} & \\ & & 1 \end{bmatrix}$$

Also note that as $J \vec{e}_i = \lambda \vec{e}_i + \vec{e}_{i-1}$,
 $(J - \lambda I) \vec{e}_i = \vec{e}_{i-1}$ and thus $(J - \lambda I)^2 \vec{e}_i = \vec{e}_{i-2}$

Examples:

i. Let $A \in \mathbb{C}^{3 \times 3}$ be such that $p = (\lambda_1 - \lambda)^3$
and $\text{nullity}(A - \lambda_1 I) = 1$ then
 $\mu = (x - \lambda_1)^{\boxed{3}}$ and A is similar to

$$J = \begin{bmatrix} \lambda_1 & 1 & \\ & \lambda_1 & 1 \\ & & \lambda_1 \end{bmatrix}$$

ii. Let $A \in \mathbb{C}^{4 \times 4}$ be such that $p = (\lambda_1 - \lambda)^4$
and $\mu = (x - \lambda_1)^2$ then $\text{nullity}(A - \lambda_1 I)$ is

$2 \text{ or } 3$ and A is similar to

$$\begin{bmatrix} \left(\begin{smallmatrix} \lambda & 1 \\ & \lambda \end{smallmatrix} \right) \\ \left(\begin{smallmatrix} \lambda & 1 \\ & \lambda \end{smallmatrix} \right) \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} \left(\begin{smallmatrix} \lambda & 1 \\ & \lambda \end{smallmatrix} \right) \\ \left(\begin{smallmatrix} \lambda & \\ & \lambda \end{smallmatrix} \right) \end{bmatrix}$$

$(4 = 2+2)$
 $(4 = 2+1+1)$

Exercises:

1) Let A be similar to the Jordan block matrix

$$J = \begin{bmatrix} \boxed{3} & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \boxed{3} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \boxed{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boxed{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \boxed{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \boxed{0} & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & \boxed{0} & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & \boxed{0} & 0 & 0 \end{bmatrix}_{8 \times 8}$$

find the characteristic and minimal polynomials of A .

$$\begin{aligned} p &= (3-\lambda)^3 (2-\lambda)^2 (-\lambda)^3 \\ \mu &= (x-3)^{\boxed{2}} (x-2)^{\boxed{1}} x^{\boxed{3}} \\ &= (x-3)^2 (x-2) x^3 \end{aligned}$$

4) let A have charact. polynomial of $p = (2-\lambda)^5 (3+\lambda)^3$ and let $\text{nullity}(A-2I) = 3$ and $\text{nullity}(A+3I) = 2$. Find all possible minimal polynomials and Jordan forms (up to reorderings)

$\lambda = 2$:

- alg mult. of 5
- geom. mult. of 3 (3 blocks)
- partitions of 5: $(3,1,1)$, $(2,2,1)$

$\lambda = -3$:

- alg mult. of 3
- geom mult. of 2
- partitions of 3: $(2,1)$

$\therefore \mu = (x-2)^\alpha (x+3)^2$ where $\alpha = 2$ or 3
and A is similar to

$$\begin{bmatrix} 2 & 1 & & & \\ & 2 & 1 & & \\ & & 2 & & \\ & & & 2 & \\ & & & & 2 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 2 & 1 & & & \\ & 2 & 1 & & \\ & & 2 & & \\ & & & 2 & \\ & & & & -3 \end{bmatrix}$$

Complex Operators

Recall: For $\vec{z} \in \mathbb{C}^n$ define $\vec{z}^*$ by

$$\vec{z}^* = \overline{\vec{z}}^T$$

$\vec{z}^H$ or $\vec{z}^+$

$$\overline{a+ib} = a-ib$$

Example:
$$\begin{bmatrix} i \\ 2-i \\ e^{i\theta} \end{bmatrix}^* = \begin{bmatrix} -i & 2+i & e^{-i\theta} \end{bmatrix}$$

Def 19.2 Let $T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^m)$ and let

$A = [T]_{\{\vec{e}_i\}}$ so that $A \in \mathbb{C}^{m \times n}$. Then the adjoint of T denoted by $T^* \in \mathcal{L}(\mathbb{C}^m, \mathbb{C}^n)$ such that for any $\vec{z} \in \mathbb{C}^n$ and $\vec{w} \in \mathbb{C}^m$

$$\langle \vec{w}, T\vec{z} \rangle_{\mathbb{C}} = \langle T^*\vec{w}, \vec{z} \rangle_{\mathbb{C}} \quad (\langle \vec{z}, \vec{w} \rangle_{\mathbb{C}} = \vec{z}^* \vec{w})$$

and has matrix representation of

$$[T^*]_{\{\vec{e}_i\}} = A^* = \overline{A}^T$$

Here $A^* = \overline{A}^T$ is said to be the adjoint or Hermitian of A .

Note that if A is real then $A^* = A^T$

Properties 19.3 : let $\vec{z}, \vec{w}, \vec{w}_i \in \mathbb{C}^n$ and $A \in \mathbb{C}^{n \times n}$

i. $\langle \vec{z}, \vec{w} \rangle = \overline{\langle \vec{w}, \vec{z} \rangle}$

ii. $\langle \vec{z}, \sum c_i \vec{w}_i \rangle = \sum c_i \langle \vec{z}, \vec{w}_i \rangle$
(linearity in the second argument)

iii. $\langle \vec{z}, A\vec{w} \rangle = \langle A^* \vec{z}, \vec{w} \rangle$

Def 19.4 A matrix $A \in \mathbb{C}^{n \times n}$ is said to be self-adjoint or Hermitian iff $A^* = A$.

Note that if A is a real, self-adjoint matrix then A is symmetric ($A = A^T$)

Example : $A = \begin{bmatrix} 5 & 3+2i \\ 3-2i & 12 \end{bmatrix}$ is self-adjoint

note that $\overline{3+2i} = 3-2i$ and $\text{diag } A$ is real

Theorem 19.5 let A be self-adjoint (i.e. A is symmetric or Hermitian), then

i. Its e-values are real

ii. Its e-vectors are orthogonal

iii. A is diagonalizable

Proof: Let $A^* = A$

i. Let $A\vec{v} = \lambda\vec{v}$ such that $\|\vec{v}\|=1$. Then

$$\begin{aligned}\lambda &= \lambda \langle \vec{v}, \vec{v} \rangle \\ &= \langle \vec{v}, \lambda \vec{v} \rangle \quad (19.3.ii) \\ &= \langle \vec{v}, A\vec{v} \rangle \\ &= \langle A\vec{v}, \vec{v} \rangle \quad A^* = A \\ &= \langle \lambda \vec{v}, \vec{v} \rangle \\ &= \overline{\langle \vec{v}, \lambda \vec{v} \rangle} \\ &= \bar{\lambda} \langle \vec{v}, \vec{v} \rangle \\ &= \bar{\lambda}\end{aligned}$$

as desired.

ii. Exercise

iii. Lemma If A is self-adjoint and λ is a (real) e-value of A , then $\text{null}(A - \lambda I) = \text{null}(A - \lambda I)^2$.

Proof: Exercise

Proof from lemma: Assume that A is not

diagonalizable so that λ is defective.

Then let $n \geq 2$ be such that

$$(A - \lambda I)^n \vec{x} = \vec{0} \text{ but } (A - \lambda I)^{n-1} \vec{x} \neq \vec{0}.$$

However, setting $\vec{y} = (A - \lambda I)^{n-2} \vec{x}$ we
 $\neq \vec{0}$

have that $\vec{y} \in \text{null}(A - \lambda I)^2$ and thus

$\vec{y} \in \text{null}(A - \lambda I)$. Now as

$$(A - \lambda I) \vec{y} = (A - \lambda I)^{n-1} \vec{x} = \vec{0}$$

we have a contradiction.

Def 19.6 A matrix U is said to be unitary
iff $U^{-1} = U^*$.

Example: The matrix $U = \frac{1}{\sqrt{2}} \begin{bmatrix} i & 1 \\ 1 & i \end{bmatrix}$

is unitary as $U^{-1} = \frac{-1}{\sqrt{2}} \begin{bmatrix} i & -1 \\ -1 & i \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -i & 1 \\ 1 & -i \end{bmatrix}$

$$\begin{bmatrix} i \\ 1 \end{bmatrix}^* \begin{bmatrix} 1 \\ i \end{bmatrix} = [-i \ 1] \begin{bmatrix} 1 \\ i \end{bmatrix} = 0 = U^*$$

Theorem 19.7 (Spectral Decomposition)

Let A be self adjoint. Then there exists

unitary matrix U and real diagonal matrix Λ such that $A = U \Lambda U^*$.

Note that this is the diagonalization of A where the e-vectors are chosen to be orthonormal

In particular if A is self-adjoint and $A \vec{u}_i = \lambda_i \vec{u}_i$ where $\{\vec{u}_i\}$ are orthonormal then

$$A = \sum_i \lambda_i \vec{u}_i \vec{u}_i^*$$

rank-1
decomp.

Exercise:

2) Find the spectral decomposition of the self-adjoint matrix

$$(a+bi)(a-bi) = a^2 + b^2$$

$$A = \begin{bmatrix} 3 & 2+i \\ 2-i & 7 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 3-\lambda & 2+i \\ 2-i & 7-\lambda \end{vmatrix}$$

$$= \lambda^2 - 10\lambda + 21 - \underbrace{(2+i)(2-i)}_5$$

$$= \lambda^2 - 10\lambda + 16 = (\lambda - 8)(\lambda - 2)$$

$$\lambda = 8, 2$$

$$\lambda_1 = 8 : A - 8I = \begin{bmatrix} -5 & 2+i \\ 2-i & -1 \end{bmatrix}, \quad \vec{v}_1 = \begin{bmatrix} 1 \\ 2-i \end{bmatrix}$$

$$\vec{u}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{1}{\sqrt{1^2 + 2^2 + 1^2}} \begin{bmatrix} 1 \\ 2-i \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2-i \end{bmatrix}$$

$$\lambda_2 = 2 : A - 2I = \begin{bmatrix} 1 & 2+i \\ 2-i & 5 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 2+i \\ -1 \end{bmatrix}$$

$$\vec{u}_2 = \frac{\vec{v}_2}{\|\vec{v}_2\|} = \frac{1}{\sqrt{6}} \begin{bmatrix} 2+i \\ -1 \end{bmatrix}$$

$$\therefore A = U \Lambda U^* \quad \text{where} \quad U = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & 2+i \\ 2-i & -1 \end{bmatrix}$$

$$\text{and} \quad \Lambda = \begin{bmatrix} 8 & \\ & 2 \end{bmatrix}$$

or

$$A = \frac{8}{6} \begin{bmatrix} 1 \\ 2-i \end{bmatrix} \begin{bmatrix} 1 & 2+i \end{bmatrix} + \frac{2}{6} \begin{bmatrix} 2+i \\ -1 \end{bmatrix} \begin{bmatrix} 2-i & -1 \end{bmatrix}$$

$$= \frac{8}{6} \begin{bmatrix} 1 & 2+i \\ 2-i & 5 \end{bmatrix} + \frac{2}{6} \begin{bmatrix} 5 & -2-i \\ -2+i & 1 \end{bmatrix}$$

Complex norm : If $\vec{z} = (z_1, \dots, z_n)$ where $z_i = a_i + ib_i$, then

$$\begin{aligned} \|\vec{z}\|_{\mathbb{C}} &= \sqrt{\langle \vec{z}, \vec{z} \rangle_{\mathbb{C}}} = \left(\sum_{i=1}^n \|z_i\|^2 \right)^{1/2} \\ &= \left(\sum_{i=1}^n a_i^2 + b_i^2 \right)^{1/2} \end{aligned}$$